# Prashant Piprotar

Senior Software Engineer I Backend

prashplus@gmail.com I +91 95746 30231 I Bengaluru I www.prashplus.com I linkedin.com/in/prashantpiprotar I github.com/prashplus

## PROFESSIONAL SUMMARY

Senior Backend Software Engineer with 5+ years of experience and an M.Tech in Computer Science from IIT Kanpur. Builds public REST APIs, third-party financing integrations, and low-latency microservices using Node.js, Python, and Java. Strong foundation in distributed systems and high-performance computing (Apache Spark, OpenMPI) with hands-on cloud experience across AWS, GCP, and Azure, Docker, and CI/CD.

## WORK EXPERIENCE

**Senior Software Engineer** 06/2023 – Present
*Enphase Energy* *Bengaluru, India*

- Designed and launched the Solargraf Public REST API and developer documentation in 4 months, reducing partner onboarding time and accelerating third-party integrations.
- Built end-to-end backend integrations with multiple Third-Party Ownership (TPO) financing providers, automating credit, lease, and purchase workflows for solar installers and homeowners.
- Defined engineering standards across the Solargraf platform by improving API performance, expanding automated test coverage, and mentoring engineers across teams.

**Software Engineer** 08/2021 – 05/2023
*Enphase Energy* *Bengaluru, India*

- Built an automated Bill of Materials (BOM) calculator with real-time pricing integration in 3 months, increasing customer satisfaction scores by 5%.
- Optimized front-end rendering and back-end database queries across core installer tools, reducing latency and shortening quote turnaround time.

**Research Intern** 01/2018 – 04/2018
*Centre for Development of Advanced Computing (C-DAC)* *Pune, India*

- Built an open-source network emulator and integrated IDE for software-defined networking (SDN), reducing setup time for researchers developing SDN applications.
- Deployed a Ceph distributed storage cluster for the SDN Online Lab, improving fault tolerance, high availability, and data retrieval speed.

## SKILLS

**Languages:** JavaScript, TypeScript, Python, Java, C++, SQL
**Frameworks & Libraries:** React.js, Redux, Next.js, Node.js, Express.js, Spring Boot, FastAPI, Material-UI
**Backend & APIs:** REST APIs, Microservices, WebSockets, API Design, Model Context Protocol (MCP), LangChain
**Databases & Caching:** MongoDB, MySQL, Redis
**Cloud & DevOps:** AWS (EC2, S3), GCP (GKE), Azure (VMs, Blob Storage), Docker, Kubernetes, CI/CD (CircleCI), Git
**Distributed Systems & HPC:** Apache Spark, OpenMPI, Ceph

## PROJECTS

**Evaluation of Spark and MPI for Scientific Data Analysis** 08/2020 – 05/2021
*IIT Kanpur (M.Tech Thesis) I Apache Spark, C++, OpenMPI*

- Benchmarked OpenMPI against Apache Spark on distributed clusters, comparing throughput on CPU-, memory-, and I/O-intensive workloads.
- Analyzed performance on large scientific datasets from astrophysics, meteorology, and combustion simulations.

**CloudMPI** 01/2020 – 05/2020
*IIT Kanpur I C++, OpenMPI, Distributed Systems*

- Designed a cloud-aware MPI communication library that reduces the impact of network latency on distributed computation in virtualized environments.
- Optimized MPI communication primitives to handle high-latency cloud interconnects efficiently.

## EDUCATION

**M. Tech. in Computer Science and Engineering** 07/2019 – 06/2021
*Indian Institute of Technology, Kanpur* *GPA: 8.6/10*

**B. Tech. in Computer Engineering** 07/2014 – 05/2018
*CSPIT, Charotar University of Science and Technology* *GPA: 9.04/10*

## AWARDS & SCHOLARSHIPS

**All India Rank 422, GATE** 01/2019